

# Yao Ming (Brian) Chan

## PERSONAL INFORMATION

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Email: brichan17@gmail.com  
Date of birth: November 17th 1999  
Place of birth: Ipoh, Malaysia  
Nationality: Australian  
Website: <https://brichan17.github.io>

## EDUCATION

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| 10/2023–Now     | <b>DPhil Mathematics</b><br>University of Oxford<br><b>Expected graduation date:</b> July 2027<br><b>Current thesis title:</b> On the Cuntz semigroups of non-simple $C^*$ -algebras<br><b>Supervisor:</b> Professor Stuart White |
| 03/2021–11/2022 | <b>MSc. Master of Science (Mathematics and Statistics)</b><br>University of Melbourne<br><b>Major:</b> Pure Mathematics<br><b>Thesis:</b> Wedge product matrices and applications<br><b>Supervisor:</b> Professor Arun Ram        |
| 03/2018–11/2020 | <b>BSc. Bachelor of Science</b><br>University of Melbourne<br><b>Major:</b> Mathematics and Statistics<br><b>Specialisation:</b> Pure Mathematics                                                                                 |

## TEACHING

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| 2025 | <b>Class tutor for Functional Analysis I B4.1</b><br><br>Organised and taught one set of classes (four classes total). Also played the role of the teaching assistant and marked the students' submitted work before each class. |
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| 2025       | <b>Teaching assistant for Functional Analysis II B4.2</b>                                                                                         |
|            | Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class. |
| 2023, 2024 | <b>Teaching assistant for Functional Analysis I B4.1</b>                                                                                          |
|            | Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class. |

## ACHIEVEMENTS

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| 2021, 2022 | <b>Mathematics and Statistics Masters Scholarship</b>                                                                                                                        |
|            | Three payments of \$2000 corresponding to the first three semesters of the Master of Science degree in recognition of consistent academic achievement throughout the degree. |
| 2018, 2019 | <b>Dean's Honours List</b>                                                                                                                                                   |
|            | Awarded for an average mark which lies in the top 3% of students in the Bachelor of Science degree.                                                                          |

## PREPRINTS AND PUBLICATIONS

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1. Y. Chan. *Wedge product matrices and orbits of principal congruence subgroups*, Lin Alg App. **696**, 2024, Pages 1-28.
2. Y. Chan. *The Cuntz semigroup of a unital graph  $C^*$ -algebra*, Currently in preparation.
3. Y. Chan, J. Curda, A. Shiner. *Classification of  $C^*$ -algebras with a unique ideal*, Currently in preparation.

## TALKS DELIVERED

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|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29/07/2026 | <b>The Cuntz semigroup of a unital graph <math>C^*</math>-algebra</b> (50 minute invited talk), UNSW, Sydney, Australia.                                     |
| 25/06/2026 | <b>The Cuntz semigroup of a unital graph <math>C^*</math>-algebra</b> (25 minute contributed talk), UK Operator Algebras Conference, Sabhal Mòr Ostaig, Isle |

of Skye.

- 08/06/2026      **The Cuntz semigroup of a unital graph  $C^*$ -algebra** (50 minute invited talk), Junior Functional Analysis Seminar, University of Oxford.
- 20/03/2026      **The Cuntz-Krieger relations in the context of the Cuntz semigroup** (15 minute contributed talk), Operator algebras in the south, Cardiff University.
- 11/09/2025      **Purely large extensions with purely infinite quotients** (20 minute contributed talk), Topological phenomena in the Cuntz semigroup, University of Oxford.
- 21/07/2025      **Cuntz semigroups and graph  $C^*$ -algebras** (lightning talk), YMC\*A, University of Southern Denmark, Odense.
- 04/07/2025      **Cuntz semigroups and graph  $C^*$ -algebras** (lightning talk), UK Operator Algebras Conference, Queen's University Belfast.
- 03/04/2025      **Nuclear dimension, extensions and graph  $C^*$ -algebras** (20 minute contributed talk), Young Functional Analysts' Workshop, University of Glasgow.
- 12/06/2024      **Nuclear dimension and graph  $C^*$ -algebras** (lightning talk), UK Operator Algebras Conference, Newcastle University, Newcastle upon Tyne.

## RESEARCH VISITS AND CONFERENCES

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- 31/08/2026 - 04/09/2026      **Cuntz semigroups and dynamics** Chalmers University of Technology, Gothenburg, Sweden.
- 29/07/2026 - 31/07/2026      **Research visit** UNSW, Sydney, Australia.
- 21/06/2026 - 26/06/2026      **UK Operator Algebras Conference**, Sabhal Mòr Ostaig, Isle of Skye, Scotland.

- 19/03/2026 - 20/03/2026 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.
- 10/09/2025 - 12/09/2025 **Topological phenomena in the Cuntz semigroup**, University of Oxford, Oxford, United Kingdom.
- 21/07/2025 - 25/07/2025 **Young Mathematicians in C\*-algebras**, University of Southern Denmark, Odense, Denmark.
- 02/07/2025 - 04/07/2025 **UK Operator Algebras Conference**, Queen's University Belfast, Belfast, Northern Ireland.
- 02/04/2025 - 04/04/2025 **Young Functional Analysts' Workshop**, University of Glasgow, Glasgow, Scotland.
- 07/03/2025 - 07/03/2025 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.
- 12/06/2024 - 14/06/2024 **UK Operator Algebras Conference**, Newcastle University, Newcastle upon Tyne, United Kingdom.
- 06/11/2023 - 10/11/2023 **Twinned Conference on C\*-algebras and Tensor Categories**, The University of Edinburgh, Edinburgh, Scotland.

## SKILLS

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Coding experience with Python

2021

### **Quantum chemistry project for the subject COMP90072**

Used NumPy to implement the Hückel method in order to model the energy levels, electron densities and bond orders of conjugated organic molecules. The Hartree-Fock method was also implemented to compute the orbital energies of small atoms with occupied 1s and 2s orbitals and diatomic molecules comprised of these atoms.

## LANGUAGES

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English (Native)

## PROFESSIONAL INVOLVEMENT

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2025-2026      **Co-organiser** for the “Young Mathematicians in  $C^*$ -algebras” conference, held at the University of Oxford in 2026.

Managed the email inbox for the conference. Responded to various queries and requests from participants.

## REFERENCES

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**Professor Arun Ram** (MSc supervisor)

Institute: University of Melbourne

Email: aram@unimelb.edu.au

**Professor Stuart White** (DPhil supervisor)

Institute: University of Oxford

Email: stuart.white@maths.ox.ac.uk